

Novel Cerium Hexacyanoferrate(II) as Cathode Material for Sodium-Ion Batteries

Jian Peng,^{†,‡} Chang Li,^{†,‡} Jinwen Yin,^{†,‡} Jinsong Wang,[‡] Yonghui Yu,[†] Yi Shen,[†] Ju Fang,[§] Annan Chen,[†] Yue Xu,[†] Ratul Rehman,[†] Chun Fang,[†] Ling Miao,[‡] Renzhi Jiang,^{||} Qing Li,^{†,§} Jiantao Han,^{*,†,||} and Yunhui Huang^{†,||}

[†]State Key Laboratory of Material Processing and Die & Mold Technology, School of Materials Science and Engineering, Huazhong University of Science and Technology, Wuhan 430074, China

[‡]School of Optical and Electronic Information, Huazhong University of Science and Technology, Wuhan 430074, China

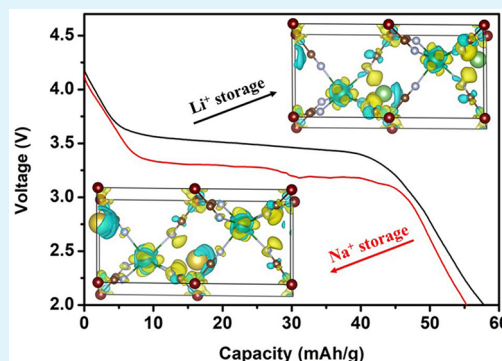
[§]Key Laboratory of Artificial Micro- and Nano-Structures of Ministry of Education, School of Physics and Technology, Wuhan University, Wuhan 430072, China

^{||}Analytical & Testing Center, Huazhong University of Science and Technology Wuhan 430074, China

Supporting Information

ABSTRACT: Owing to the demand for large-scale energy storage applications and the Li resource shortage, over the past few decades, various novel materials have been investigated as Li⁺-ion hosts. However, very few candidates have been found as Na⁺/K⁺-ion host due to their much larger ionic radius than that of Li⁺-ions. Herein, a cerium hexacyanoferrate (CeHCF) has been successfully synthesized by a facile coprecipitation method for Li⁺/Na⁺/K⁺-ion storage. As a cathode material, CeHCF delivers a specific capacity of ~60 mAh/g at 0.25 C and ~80% capacity retention at 8.3 C in lithium-ion batteries (LIBs), and ~55 mAh/g at 0.25 C with ~55% capacity retention at 8.3 C in sodium-ion batteries (SIBs), respectively. Owing to its open 3D framework and highly stable crystal structure, CeHCF presents as a “zero-strain” insertion cathode material for a reversible insertion/extraction of Li⁺/Na⁺/K⁺-ion. Through multiple characterization techniques, including *in situ* XRD and *ex situ* Raman, alkali ion intercalation of CeHCF has been investigated. Our studies not only deliver a novel candidate as a universal host for alkali ion but also deduce the alkali ion storage mechanism in the framework and the structure evolution in the electrochemical process.

KEYWORDS: Prussian blue analogues, alkali ion batteries, cerium hexacyanoferrate, sodium-ion batteries, rare earth metals, universal host



With increasing energy consumption, low-cost and long-life rechargeable batteries are urgently demanded for both portable devices and large-scale energy storage systems.^{1,2} Monovalent alkali metal ions are of great importance as energy carriers for practical applications; in particular, lithium-ion batteries (LIBs) have been extensively studied and widely applied owing to their merits of environmental benignity, high energy density, and long cycle life.³ However, due to the shortage of lithium resources, the large-scale commercial applications of LIBs face serious issues in the future.⁴ Sodium and potassium, in contrast, are abundant and easy accessible; both of these are suitable candidates employed in rechargeable batteries.^{5,6} Over the past few years, sodium-ion batteries (SIBs) and potassium-ion batteries (KIBs) have been widely studied for potential large-scale stationary applications.^{7,6} However, SIBs and KIBs always suffer from low specific capacity and poor cycle life, probably due to the larger ionic radii of Na⁺ (1.02 Å) and K⁺ (1.38 Å) than that of Li⁺ (0.76 Å).⁸

Prussian blue analogues (PBAs) are a large family of framework materials with the chemical formula A_xM[Fe(CN)₆]_{1-y}□_y·zH₂O (where A is alkali metal, M is transition metal, □ represents vacancies) and have attracted increasing attention for developing novel energy storage systems.^{9–11} PBAs share a metal–ligand open framework, and their cavities provide massive interstitial sites for alkali/alkaline-earth ion storage.^{12–16,10} Recently, Chen et al. successfully prepared an indium hexacyanoferrate as cathode material with high voltage and fast kinetics;¹⁷ Lee et al. fabricated high-performance aqueous batteries based on vanadium hexacyanoferrate.¹⁸ In addition, Wu et al. reported an interesting rocking-chair NH₄⁺-ion battery based on (NH₄)_{1.47}Ni[Fe(CN)₆]_{0.88}.¹⁹

In this study, a new member of Prussian blue analogues (PBAs), cerium hexacyanoferrate (CeHCF), was successfully

Received: October 3, 2018

Accepted: December 10, 2018

Published: December 10, 2018

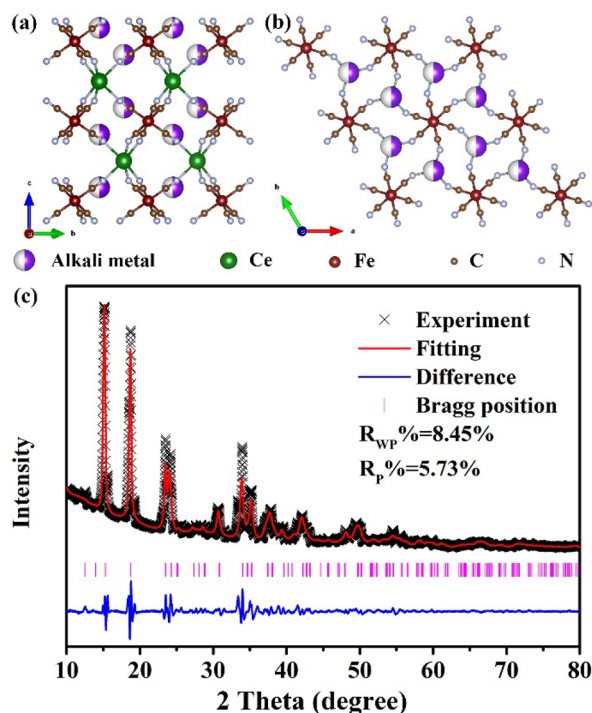


Figure 1. Schematic crystal structure of CeHCF frameworks: (a, b) local structures of Ce-HCF and (c) X-ray diffraction (XRD) Rietveld refinement pattern.

obtained by a convenient and simple method, serving as “zero-strain” cathode material for alkali ion (Li^+ , Na^+ , K^+) storage. A density-functional theory (DFT) calculation was carried out to identify the framework structure evolution and storage mechanism during charge/discharge processes. Our results not only provide a promising candidate for developing high-performance rechargeable batteries but also bring insight into the structure–property relationship of CeHCF by experimental tests and theoretical simulations.

The sample crystal structures were investigated by X-ray diffraction (XRD) and refinements by the Rietveld method in GASA software. Figure 1c shows that CeHCF is hexagonal (space group = $P6_3/m$, $a = 7.3789 \text{ \AA}$, $c = 14.0601 \text{ \AA}$) with good crystallinity. The refined local structures are also illustrated in Figure 1a,b; CeN_6 and FeC_6 polyhedra in the host framework are bridged by $\text{C}\equiv\text{N}$ units, and the guest alkali ion inserts into the pores of the host framework. The $\text{Fe}-\text{C}\equiv\text{N}-\text{Ce}$ linkage of CeHCF is nearly linear (the angles of $\text{Ce}-\text{N}-\text{C}$ and $\text{Fe}-\text{C}-\text{N}$ are 156.07° and 180° , respectively), which is the same as that of the reported rhombohedral structured PBAs.^{20,21} According to the results of refinement and TG, the chemical formula of CeHCF is $\text{Na}_{0.53}\text{Ce}[\text{Fe}(\text{CN})_6]_{0.77}\cdot 4.12\text{H}_2\text{O}$. The large internal spaces ensure CeHCF to act as a universal ion host. The FT-IR spectra (Figure S1) of CeHCF exhibit typical features of PBAs. The water contents (including absorbed water and crystallized water) and vacancy in the framework of CeHCF result in a slight different absorption band of the water molecules.²² The adsorption peak located at 2088 cm^{-1} belongs to the $\text{C}\equiv\text{N}$ group stretch mode, which is coordinated with the inner Fe^{2+} ($\text{Fe}^{2+} \leftarrow \text{C}$), and the peak located at 595 cm^{-1} can be ascribed to $\text{Fe}-\text{C}$. The sharp absorptions located at 1621 and 3552 cm^{-1} are associated with the $\text{H}-\text{O}$ stretching and $\text{H}-\text{O}-\text{H}$ bending modes, arising from the

lattice water, while the absorption peak located at 3617 cm^{-1} is arising from the free surface water.²³

The morphology of the electrode material can be regarded as a key factor relating to electrochemical performance in rechargeable batteries.^{24,25} Typically, nanosized and uniform particles may increase the contact area between electrode and electrolyte.^{10,26,27} The uniform dispersion of Ce, Fe, and N elements in CeHCF is confirmed by energy dispersive spectroscopy (EDS) mapping images in Figure S3, which also suggests that these samples have high purity and integrity.²⁸ According to the high-resolution transmission electron microscopy (HRTEM) in Figure S5, the obvious lattice spacings of 0.404 and 0.455 nm are shown by the clear lattice fringes, which are consistent with the (110) and (012) lattice distances, respectively.

To explore the potential application of CeHCF as cathode material for rechargeable Li^+/Na^+ -ion batteries, the electrochemical performances associated with Li^+/Na^+ insertion/extraction were investigated. Figure S6a shows the first cyclic voltammogram (CV) of CeHCF at a scan rate of 0.1 mV s^{-1} within 2.0 – 4.2 V (vs Li^+/Li). CeHCF appears as a pair of well-defined reduction/oxidation peaks located at 3.48 V , which corresponds to the equilibrium potential of $\text{Fe}^{2+}/\text{Fe}^{3+}$. In accord with the CV curves, the galvanostatic charge/discharge voltage profiles in the half-cell at 0.1 C exhibit distinguishable potential plateaus at 3.5 V , and the initial charge/discharge capacities are 61.4 and 57.9 mAh g^{-1} , respectively, corresponding to a Coulombic efficiency of 94.3% . Part of the capacity loss is mainly caused by formation of a solid electrolyte interphase and irreversible side-reaction between electrode material and electrolyte.¹⁸ As displayed in Figure 2c,e, CeHCF exhibits a superhigh rate performance, from 20 to 500 mA g^{-1} . The CeHCF cathode shows a capacity retention of 77% , indicating an excellent rate performance. In addition, the long-term cycle performance is shown in Figure 2f; after activation under 20 mA g^{-1} for 5 cycles, CeHCF shows a negligible capacity fading and voltage polarization increase after 80 cycles at a current density of 150 mA g^{-1} ($\sim 2.5 \text{ C}$) with high Coulombic efficiencies (CEs) of $\sim 98\%$ after an activated process.

Figure 2b,d shows that Na-ion storage behaviors of CeHCF are similar to those of Li-ion storage with larger polarization which is due to the larger ionic radius of Na^+ than that of Li^+ . Also, in order to avoid the side-reaction as shown in Figure 2b, the charge and discharge window was adjusted to 2.0 – 4.0 V . The charge and discharge capacities of the first cycle reach 62.5 and 55.7 mAh g^{-1} , respectively, corresponding to a CE of $\sim 90\%$ and increasing in the following cycles. Significantly, CeHCF demonstrates a remarkable cycling stability compared to those of other members of the PBA family.¹⁰ At the current density of 150 mA g^{-1} , its specific capacity reaches 52.7 mAh g^{-1} with a high CE, nearly 100% , during the entire long-term cycling. The high CE, low cost, and compatibility of CeHCF for Na^+ -ion make its products very attractive in a large-scale energy storage system.^{21,29–32}

The structure evolution during Na^+ -ion insertion/extraction processes was investigated through *in situ* XRD studies. Figure 3 shows that, during the Na^+ extraction process, remarkably, the peaks at $\sim 23.5^\circ$ and $\sim 39^\circ$ shift slightly to the low-degree direction while the main peak at $\sim 14^\circ$ of CeHCF remains unchanged, indicating the stable crystal structure of CeHCF and without any structural phase transition during the charging process. In addition, as shown in Figure S7, the $\text{C}-\text{N}$ bond

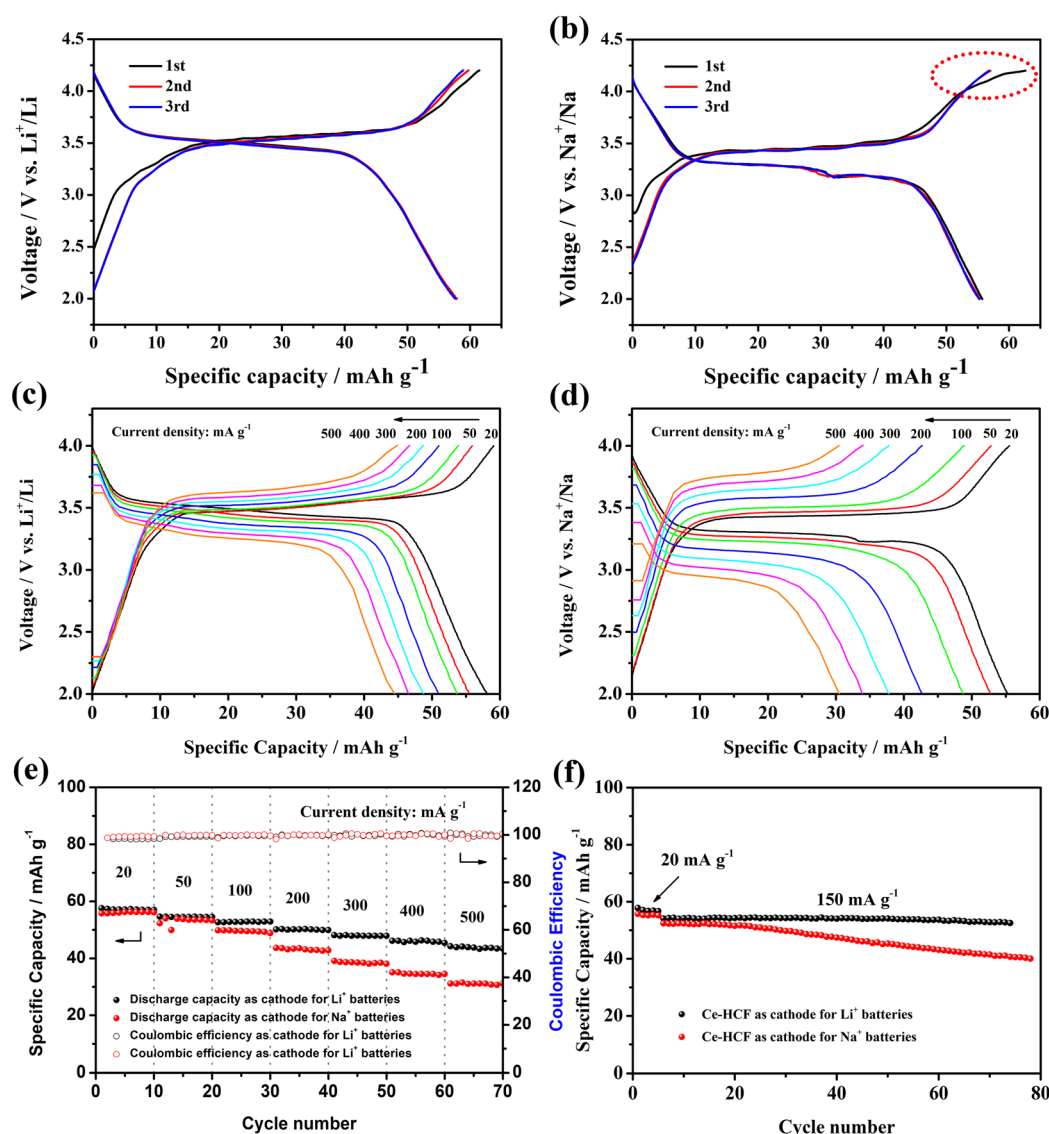


Figure 2. Electrochemical properties of CeHCF. (a, b) Galvanostatic charge/discharge voltage profiles of CeHCF as cathode materials for Li⁺ and Na⁺ batteries, respectively. (c, d) CeHCF as cathode materials for charge/discharge at different current densities. (e, f) Rate and cycle performance of CeHCF as cathode for Li⁺ and Na⁺ batteries.

remains unchanged during the charge and discharge and the peak shift is negligible during the charging process (Na⁺ extraction suggests that the lattice parameters are quite stable, leading to highly reversible Na⁺ insertion/extraction processes and further avoiding the crystal framework collapse). During the discharge process to 2.0 V (corresponding to Na⁺ insertion), the XRD pattern of CeHCF shifts to its initial position after the full insertion of Na⁺, indicating the high reversible structural evolution during the charge/discharge processes. The extraction voltage of M ions (Li⁺, Na⁺, and K⁺) can be calculated according to the following equation:

$$V = -(E_2 - E_1 - E_M)/e \quad (1)$$

Here, E_2 , E_1 , and E_M represent the energy levels of $M_2CeFe(CN)_6$, $M_1CeFe(CN)_6$, and metallic M, respectively. Table S1 shows that the voltages were calculated with eq 1 for the extraction of Na⁺-ion. The results show that the extraction voltages of 3.35 and 3.28 V for Li⁺ and Na⁺, respectively, are very consistent with the oxidation peaks at 3.59 and 3.55 V. The difference of charge density (as shown in Figure 4) shows

that Li is positively charged, and electrons transfer to near C or N, which causes higher extraction voltages compared with that of Na. Furthermore, the theoretical voltage of K⁺ insertion/extraction is ~3.76 V, indicating CeHCF could serve as K⁺ host besides Li⁺ and Na⁺.

In summary, we reported a new member of the PBA family, CeHCF, as a universal host material for alkali ions. Due to an open 3D network and large lattice space, the hexagonal CeHCF cathode exhibits a specific capacity of ~55 mAh g⁻¹ with an ultrahigh rate capability. Furthermore, the potential application as cathode material for KIBs is validated through DFT calculation, thereby offering considerable potential for wide use in rechargeable batteries. This proof-of-concept rare earth metal PBA cathode material CeHCF is competitive compared with most of the well-known cathode materials for large-scale energy storage systems.

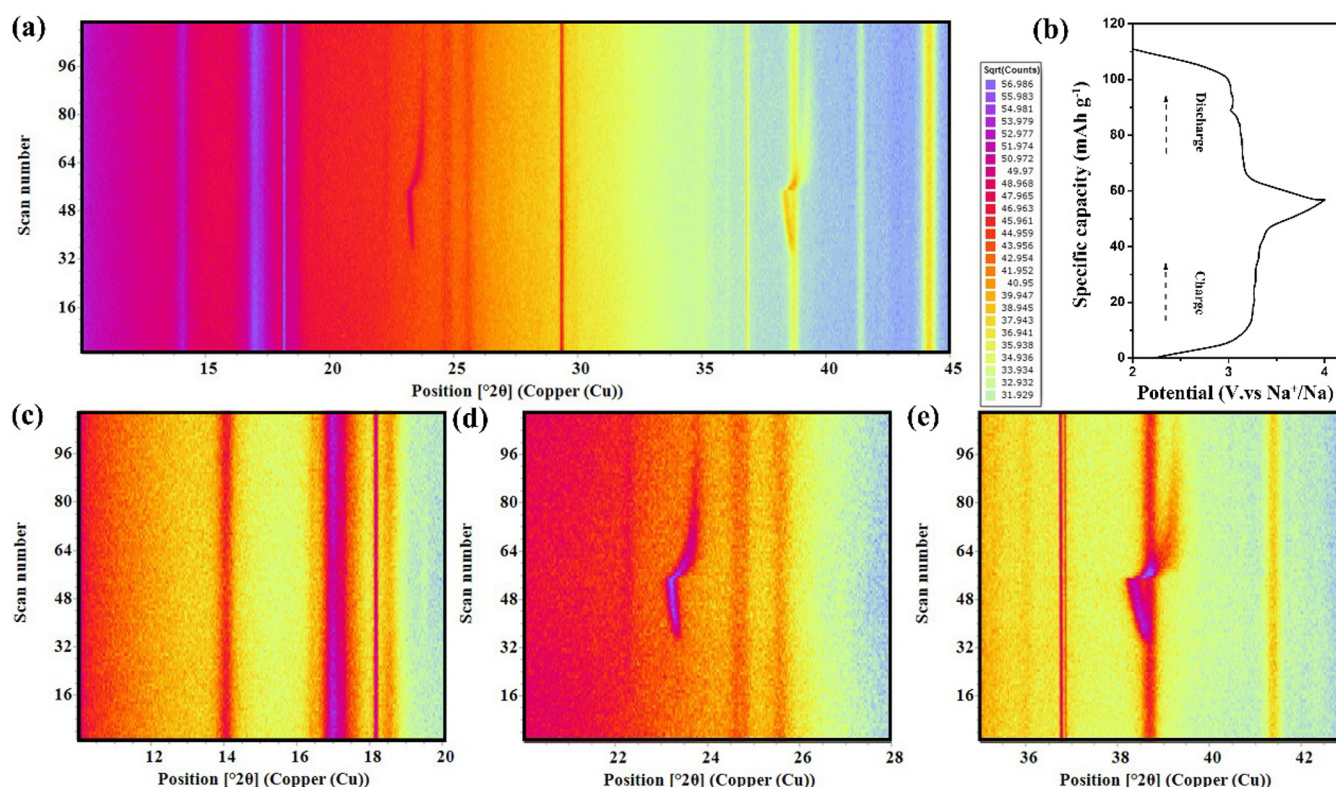


Figure 3. *In situ* XRD patterns of CeHCF under cycling at a current density of 6 mA g^{-1} . (a) Contour maps of *in situ* XRD during the first cycle. (b) Galvanostatic charge/discharge voltage profiles at a current density of 6 mA g^{-1} . (c–e) The image plots of $10\text{--}20^\circ$, $20\text{--}28^\circ$, and $35\text{--}43^\circ$, respectively.

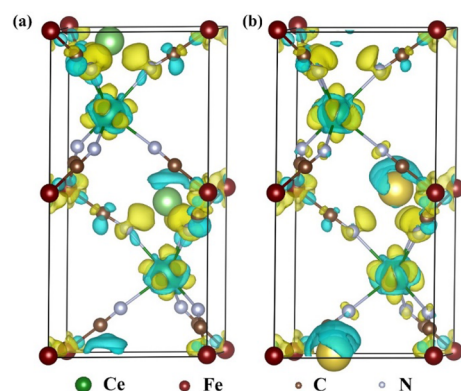


Figure 4. DOS patterns of Ce-HCF insert (a) Li^+ and (b) Na^+ , respectively.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: [10.1021/acsaem.8b01686](https://doi.org/10.1021/acsaem.8b01686).

IR, TG, SEM, TEM, CV, *ex situ* Raman, and *in situ* XRD (PDF)

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: jthan@hust.edu.cn (J.H.).

ORCID

Qing Li: 0000-0003-4807-030X

Jiantao Han: 0000-0002-1302-7136

Yunhui Huang: 0000-0003-1687-1938

Author Contributions

[†]J.P., C.L., and J.Y. contributed equally to this work. The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the National Key R&D Program of China (Grant 2016YFB0100302), and the National Natural Science Foundation of China (Grants 51772117 and 51732005). The authors also thank the Analytical and Testing Centre of HUST and the State Key Laboratory of Materials Processing and Die & Mold Technology of HUST for XRD, SEM, TEM, Raman, TGA, and other measurements.

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